18 At the start of year n, the population of a village is V_n

At the start of the following year, the population of the village is V_{n+1} where

$$V_{n+1} = 0.8\,V_n$$

At the start of 2018 the population of the village was 375

Show that the population of the village was less than 200 at the start of 2021

Start of 2018: 375

Start of 2019: 375 × 0.8 = 300

Start of 2020: 300 × 0.8 = 240

Start of 2021: 240 × 0.8 = 192

192 < 200, as required.

1 mark
Apply the recurrence.

1 mark
Three annual steps from 2018 to 2021.

1 mark
State the calculated value and compare it with 200.

(Total for Question 18 is 3 marks)

DO NOT WRITE IN THIS AREA
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